Complete Java: From Basics to Advanced System Design

1. Java Basics (Core Language)

1. Variables (primitive & reference types)
2. Data types, type casting
3. Operators (arithmetic, logical, relational)
4. Control flow (if, switch, loops)
5. Methods and parameter passing
6. Arrays and basic data handling

2. Classes & Object-Oriented Programming

1. Class and object creation
2. Constructors and initialization
3. Encapsulation, inheritance, polymorphism, abstraction
4. Access modifiers (public, private, protected, default)
5. Method overloading & overriding
6. Interfaces and abstract classes

3. Java Memory Model & JVM Internals

1. Heap, Stack, Metaspace
2. Object lifecycle and allocation
3. Garbage Collection (G1, ZGC basics)
4. Class loading (Bootstrap, Application loaders)
5. JVM tuning and memory leaks

4. Collections & Data Structures

1. List, Set, Map internals
2. HashMap, ConcurrentHashMap internals
3. Tree structures and graphs
4. Sorting, searching algorithms
5. Time & space complexity

5. Concurrency & Multithreading

1. Thread lifecycle and creation
2. Synchronization, volatile, locks
3. ExecutorService, CompletableFuture
4. Deadlocks, race conditions
5. Java Memory Model (happens-before)

6. I/O, NIO & Networking

1. InputStream, OutputStream
2. File handling
3. Sockets and networking basics
4. NIO (Channels, Buffers, Selectors)
5. Non-blocking I/O and reactive systems

7. Functional Programming

1. Lambdas and functional interfaces
2. Streams API
3. Optional usage
4. Parallel streams

8. Exception Handling

1. Checked vs unchecked exceptions
2. Custom exceptions
3. Best practices and resilience patterns

9. Design Principles & Patterns

1. SOLID principles
2. Design patterns (Factory, Singleton, Strategy, Observer)
3. Clean code and modular design

10. Performance & Tuning

1. CPU and memory profiling

2 Garbage collection tuning
3 Thread pool tuning
4 Latency vs throughput

11. Security Practices

1 Input validation
2 Secure coding practices
3 Serialization risks
4 Authentication basics

12. Distributed Systems Concepts

1 Partitioning and hashing
2 Replication
3 CAP theorem
4 Event-driven architecture
5 Caching strategies